

Critical Analysis and Comparison of Autoencoders for Pre-training in the Graph Domain

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Abstract

Pre-training has emerged as an important research direction in the graph domain, as it helps address several common challenges in real-world graph data, such as label scarcity and noisy data. It is also highly relevant in the context of the trend towards larger and more general graph foundation models. While contrastive pre-training has long been the dominant approach, it typically requires negative sampling and data augmentation, which is a significant disadvantage. Graph autoencoders offer a self-supervised learning approach that yields increasingly strong performance while avoiding the pitfalls of graph contrastive learning and have therefore become a highly popular alternative. This seminar paper presents an overview of the field of graph autoencoders by introducing the most popular methods and classifying them by their reconstruction target. It also provides a collection of link prediction and node classification benchmark results for in-domain pre-training, as well as node classification results for cross-domain pre-training. The methods are compared and critically discussed, particularly in the context of the trend towards larger, more general models.

CCS Concepts

• **Computing methodologies** → **Learning latent representations; Unsupervised learning; Neural networks.**

Keywords

graph neural networks, masked autoencoders, self-supervised learning

1 Introduction

Following the remarkable success of large-scale foundation models in natural language processing and computer vision, there is growing interest in transferring these achievements to the graph domain [27]. Pre-training has proven essential for effectively training such large general-purpose models [12]. This paradigm could be particularly beneficial for graph-structured data, where labels are frequently sparse, expensive to obtain, and susceptible to noise [6].

There is a wide variety of pre-training methods for graph neural networks that can be broadly categorized into supervised and self-supervised approaches [12]. While supervised methods demonstrate high efficacy for specific, narrowly defined tasks, their practical applicability remains limited [27]. Graph labels are often prohibitively expensive to obtain and require domain-specific expertise [27]. Furthermore, these approaches suffer from limited scalability and generalizability, achieving effectiveness primarily when the pre-training task closely aligns with the downstream application [25].

Consequently, self-supervised approaches have emerged as the dominant paradigm for pre-training in recent years [4]. These methods can be further subdivided into contrastive and generative techniques [4].

For an extended period, contrastive methods represented the most successful approach in this domain [4]. These methods typically learn representations by maximizing agreement between differently augmented views of the same graph while minimizing agreement with views from different graphs [21]. Despite demonstrating strong performance, contrastive methods suffer from several significant limitations. First, they generally require negative sampling, where the construction of effective negative samples constitutes a computationally expensive and often graph-dependent challenge [4]. Second, these methods exhibit high sensitivity to the choice of data augmentation strategies. This dependency is problematic because graph augmentation typically relies on domain-specific heuristics and lacks clear interpretability in graph-structured data [4].

Due to these limitations, generative methods have gained increasing prominence in recent years. While autoregressive approaches, which have achieved remarkable success in domains such as natural language processing, exist for graphs (e.g., GPT-GNN [7]), they are fundamentally ill-suited to graph structures due to the absence of an inherent node ordering [4].

For these reasons, autoencoders have become the predominant generative paradigm for graph pre-training. Although they historically lagged behind contrastive methods in performance, recent advances have enabled them to achieve competitive results while circumventing the need for negative sampling and data augmentation.

This seminar paper provides:

- An overview of influential and effective graph autoencoder frameworks. Given the substantial recent research activity in this area, a taxonomy organized by reconstruction target is presented. The methods are categorized into structural reconstruction, node-feature reconstruction, and hybrid approaches.
- A compilation of benchmark results for the presented methods on link prediction and node classification tasks in an in-domain setting, and node classification in a cross-domain setting.
- A critical analysis and comparison of the surveyed frameworks, along with an outlook on promising future research directions.

2 Preliminaries

2.1 Notation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote a graph, where $\mathcal{V} = \{v_1, v_2, \dots, v_n\}$ is the set of $n = |\mathcal{V}|$ nodes and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges. The structure of the graph is represented by an adjacency matrix $A \in \{0, 1\}^{n \times n}$. Since the methods discussed in this paper assume unweighted graphs, the elements of the adjacency matrix are defined as:

$$A_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in \mathcal{E} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Furthermore, let $X \in \mathbb{R}^{n \times d}$ denote the node feature matrix, where d is the input feature dimensionality, and let D denote the degree matrix of $\mathcal{G}$.

2.2 Graph Autoencoders

The primary goal of pre-training on graphs is to learn an encoder f_E that maps nodes into low-dimensional latent representations z_i , summarized in the matrix $Z \in \mathbb{R}^{n \times d_h}$, where d_h is the dimensionality of the latent space. These representations should ideally capture the essential structural and attribute-based information of their respective nodes [9].

From the perspective of Graph Autoencoders (GAEs), the model consists of the encoder f_E and a decoder f_D . To account for both traditional and masked GAEs, the encoding and reconstruction process is defined as:

$$Z = f_E(A', X'), \quad \hat{Y} = f_D(Z, \tilde{A}) \quad (2)$$

Here, A' and X' represent the input adjacency and feature matrices, which may be either the original data or modified (e.g., masked) versions. The decoder f_D produces a reconstruction $\hat{Y}$ based on the latent representation Z and, in some architectures, auxiliary structural information $\tilde{A}$ (such as the unaltered or masked adjacency matrix).

The reconstruction target $\hat{Y}$ varies depending on the pre-training objective:

- **Feature Reconstruction:** In approaches such as GraphMAE [3, 4], the target is typically the original node feature matrix X .
- **Structural Reconstruction:** Other methods focus on reconstructing structural properties, such as the adjacency matrix A [8, 9, 20], the degree matrix D [9], or specific topological features like masking bandwidths [31].
- **Hybrid Reconstruction:** Some recent approaches focus on reconstructing both structural information and node features [10, 18].

3 Overview of Graph Autoencoders

The field of GAEs has experienced a significant surge in research interest, driven by the state-of-the-art performance of recent frameworks such as GraphMAE [3, 4] and S2GAE [20]. This section provides a systematic overview of the most influential methods in the domain, tracing their underlying motivations and developmental trajectories.

Given the sheer number and diversity of recent architectures, a formal taxonomy is essential for navigating the current landscape. As mentioned earlier, one effective approach for categorizing existing methods is based on the decoder’s reconstruction target [31]. This classification is particularly useful as it reflects the historical evolution of the field, distinguishes the specific strengths of various models, and clarifies their applicability to different downstream tasks.

3.1 Structural Reconstruction

Kipf and Welling [8] pioneered this category with the Variational Graph Auto-Encoder (VGAE) and its non-probabilistic variant (GAE) [8]. The model utilizes a GNN-based encoder to map nodes into a latent space and a simple inner product decoder to reconstruct the adjacency matrix. In VGAE, the encoder parameterizes a Gaussian distribution, and the framework is trained by maximizing the variational lower bound. The reconstruction accuracy is balanced with a KL-divergence regularizer.

While demonstrating good performance on link prediction tasks, traditional structure reconstruction methods like VGAE failed to generalize onto other tasks such as node- or graph-level tasks [9]. To address this issue, MaskGAE [9] and S2GAE [20] introduced masked autoencoders to the GAEs utilizing a structural reconstruction objective. Masked autoencoding has achieved significant success in other domains such as computer vision and NLP, and is naturally applicable to the graph domain because edges can be straightforwardly masked [9]. Introducing masking addresses several limitations of earlier structure reconstruction methods:

- **Proximity Over-emphasis:** Traditional GAEs often over-emphasize local proximity at the expense of global structural information, which limits their performance on downstream classification tasks [9, 20]. Masking reduces the redundancy between the neighborhoods of adjacent nodes, forcing the model to learn more robust structural features [9].
- **Overfitting and Trivial Solutions:** Graph reconstruction without corruption risks learning trivial identity mappings [20]. Since GNNs are inherently expressive under the homophily hypothesis, GAEs without a corruption mechanism are at risk of learning trivial identity mappings rather than meaningful representations [20].

MaskGAE implements several measures to balance proximity and structural information. It introduces two primary masking strategies: edge-wise random masking and path-wise random masking [9]. The latter uses random walks to mask contiguous sequences of edges, breaking short-range connections and forcing the model to look elsewhere in the graph to infer the missing structure [9].

To improve efficiency and scalability the encoder only processes the unmasked, visible subgraph [9]. Furthermore, MaskGAE utilizes a dual-decoder strategy consisting of a structure decoder to reconstruct edges and an auxiliary degree decoder [9]. The degree decoder is tasked to predict the node degrees from the latent representations. This is a further measure against the overemphasis of proximity information and forces the encoder to capture more meaningful structural information of a node [9]. Both decoders utilize MLPs. The entire framework is depicted in Figure 1.

The concurrently developed S2GAE [20] introduces direction-aware graph masking. This strategy offers two modes: undirected masking, which is effective for dense graphs with high redundancy, and directed masking, which provides more challenging self-supervisory signals for sparse graphs [20]. Like MaskGAE, the S2GAE encoder operates only on the partial graph structure to improve generalizability [20].

A core contribution of S2GAE is its cross-correlation decoder, which addresses the limitations of standard MLP-based decoders [20] as used in MaskGAE. S2GAE argues that heavy edge masking makes latent representations noisy and susceptible to over-smoothing. Therefore, simple MLPs struggle to reconstruct the topology accurately [20]. The cross-correlation decoder explicitly captures correlations between end nodes across multiple granularities by computing element-wise products of hidden representations from different GNN layers:

$$h_{e_{vu}} = \{h_v^k \odot h_u^j\} \text{ for } k, j \in \{1, \dots, K\} \quad (3)$$

This allows the model to filter out noise and focus on informative structural motifs shared between nodes [20].

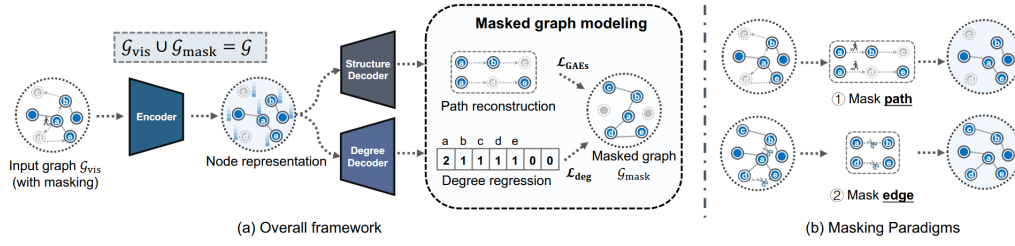


Figure 1: The Mask GAE-Framework. Source: Li et al. [9]

While discrete masking has improved the generalizability of GAEs, relying on binary edge removal can introduce new bottlenecks, such as blocking essential message flows and making deeper GNNs vulnerable to over-smoothing [31]. Recent work suggests that softening these masks into continuous values could be beneficial [31]. The authors observe that binary edge removal often blocks long-range pathways, causing performance to drop significantly when using GNNs with five or more layers [31].

To resolve this, Bandana introduces non-discrete bandwidth masks [31]. Instead of deleting an edge entirely, a bandwidth value (sampled from a continuous Boltzmann-Gibbs distribution) is assigned to each edge to restrict the amount of information passed during message propagation [31]. This ensures that the graph topology remains intact, keeping the message pathways open even under high mask ratios [31].

The framework’s primary contributions are:

- **Bandwidth Prediction:** The reconstruction objective is shifted from binary link existence to a regression task where the decoder must predict the specific continuous bandwidth values of the edges [31]. This task is more challenging and forces the model to learn a deeper understanding of the topological manifold [31].
- **Layer-wise Prediction:** Bandana generates independent bandwidth masks for every GNN layer and calculates the reconstruction loss as an average across all layers [31]. This allows the model to capture structural information at multiple granularities [31].

Theoretically, Bandana is framed as a regularized denoising autoencoder [31]. Predicting these non-discrete bandwidths is mathematically equivalent to optimizing the score in the topological encoding space, allowing Bandana to significantly outperform discrete structure reconstruction methods on both link prediction and node classification [31].

3.2 Node-Feature Reconstruction

While GAEs with structure reconstruction objectives show good results in link prediction, their performance on node and graph classification tasks has long been overshadowed by GCL [4]. To address this, GraphMAE [4] introduced a feature reconstruction-based framework. Instead of structural information, the decoder’s reconstruction objective is the node-feature matrix X of the graph. Motivated by a study showing that simple MLPs distilled from GNN teachers can achieve performance comparable to GNNs on node classification tasks [29], GraphMAE argued that feature reconstruction alone may be sufficient to learn meaningful representations for node and graph classification tasks. While GraphMAE was not the first to introduce node feature-based reconstruction (e.g., MGAE [22]), it was the first node feature

reconstruction-based method to consistently outperform state-of-the-art contrastive approaches in node and graph classification [4], sparking a lot of research in this direction. This was achieved by the following additions, which address several limitations of earlier node-feature reconstruction methods [4]:

- **Masked feature reconstruction:** GraphMAE masks a randomly sampled subset of nodes (sampled from a uniform distribution without replacement) using a learnable mask token $x_{[M]}$. This approach mitigates the risk inherent in feature-reconstruction methods without corruption, where the encoder may learn a trivial identity mapping.
- **Scaled cosine error:** GraphMAE employs a scaled cosine error (SCE) instead of the commonly used mean squared error (MSE) for calculating reconstruction loss. The authors identify two key disadvantages of MSE: sensitivity and selectivity. Sensitivity refers to MSE’s susceptibility to variations in vector norms and dimensionality, while selectivity describes MSE’s limited ability to focus on hard samples. SCE addresses the sensitivity issue by using cosine similarity, which is inherently normalized. To tackle the selectivity problem, the cosine error is scaled to reduce the weight of easy samples, thereby emphasizing the learning signal from more challenging examples.
- **Re-mask decoding:** Before the decoder reconstructs the feature matrix X from the latent code H , a uniformly sampled subset of nodes is masked again using a learnable mask token $h_{[M]}$. This further encourages the encoder to learn more compressed representations. Since neighboring nodes may be masked, the encoder is forced to rely more on structural information rather than purely on neighborhood features. Additionally, this masking serves as a form of regularization, improving robustness to noisy input data.
- **GNN decoder:** GraphMAE utilizes a GNN-based decoder, as opposed to the simple MLP or dot-product decoders employed by earlier methods. This design choice forces the encoder to learn latent representations with richer structural information by shifting the reconstruction task from individual node recovery to neighborhood-based recovery.

Figure 2 depicts the GraphMAE framework.

One of GraphMAE’s major limitations is that performance heavily diminishes when features have low discriminability [3]. This occurs because the model may fit inaccurate targets and noise, causing it to learn less effective representations. This is a serious disadvantage in the graph domain, because real-world graph data is often noisy, and node features in particular are

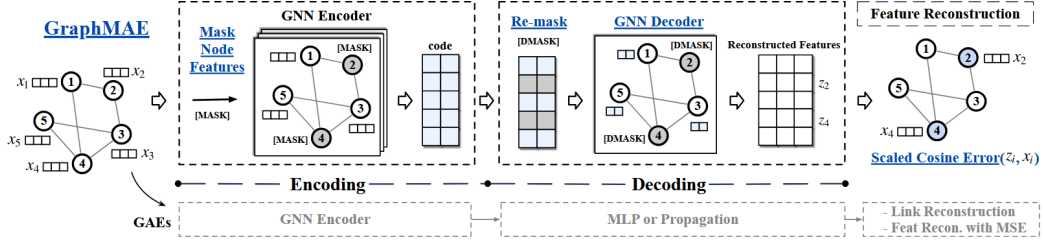


Figure 2: The GraphMAE-Framework. Source: Hou et al. [4]

oftentimes noisy approximations generated by feature extractors [3]. GraphMAE2 [3], the direct successor of GraphMAE, addresses this limitation by adding regularization to the target reconstruction. This was achieved by expanding the GraphMAE method with two additions [3]:

- **Multi-view random re-mask decoding:** The latent representation is corrupted multiple times by randomly masking different nodes each time. Each corrupted version is decoded separately to reconstruct the original features, and the model learns by comparing all reconstructions against the input. This multi-view approach acts as regularization: by forcing the model to handle multiple different corruption patterns, it becomes more robust and less likely to overfit to noise or memorize spurious patterns in the features.
- **Latent representation prediction:** Rather than only reconstructing raw features which may be noisy, the model also learns to predict a cleaner version of the latent representation. This target is generated by a separate teacher network that processes the uncorrupted graph, and the teacher’s weights are slowly updated to track the main encoder via exponential moving average. By predicting this smoother target instead of noisy inputs, the model learns more meaningful, high-level representations that capture semantic structure rather than surface-level noise.

The ablation study showed that both of these additions lead to performance gains, particularly on noisy graphs, for node classification tasks. GraphMAE2 also addresses the problem of extending the framework to very large graphs. This is outside the scope of this seminar paper, however, and will not be discussed.

The GraphMAE methods empirically demonstrate strong results on both node and graph classification tasks, and their individual components are generally well-motivated empirically. However, the methods lack theoretical understanding [23].

AUG-MAE [23] addresses this gap by proving that GraphMAE implicitly performs graph contrastive learning, which explains its competitive performance relative to contrastive approaches. Building on this theoretical foundation, AUG-MAE proposes an improved framework that explicitly considers alignment and uniformity, which are central properties of representation learning in contrastive learning [24].

This is achieved by introducing the following additions to the GraphMAE framework [23]:

- **GNN-based mask generator:** To address alignment challenges, a GNN-based mask generator is introduced. The mask generator takes the unmasked graph as input and outputs a probability for each node to be masked. A ratio regularizer maintains an appropriate masking ratio,

which is a dataset-dependent hyperparameter. The mask generator’s weights are trained adversarially against the GraphMAE parameters, enabling it to gradually learn to generate hard-to-align positive pairs while GraphMAE learns to align them.

- **Explicit uniformity regularizer:** While GraphMAE architecturally avoids full feature collapse of the latent space, the AUG-MAE paper demonstrated that partial dimensional collapse still occurs, indicating room for improved uniformity. To address this, AUG-MAE introduces an explicit uniformity regularizer that ensures the learned representations are uniformly distributed on the unit hypersphere.
- **Easy-to-hard training:** To implement an easy-to-hard training strategy, the masking protocol combines both random masking and the GNN-based mask generator. At the start of training, random sampling is predominantly used to initialize the model with easy-to-align samples. During training, the weight of the mask generator is progressively increased, forcing the GraphMAE model to improve by aligning increasingly hard-to-align pairs.

The ablation study shows that these additions are not only theoretically well-motivated but also lead to significant performance improvements on node and graph classification tasks.

3.3 Hybrid Approaches

While structural reconstruction objectives produce representations with higher structural informativeness, methods using feature reconstruction objectives generally outperform them on node- and graph-level classification tasks [18]. Conversely, structural reconstruction approaches excel at structural tasks such as link prediction [18]. This performance gap has motivated the development of hybrid methods that reconstruct both structural and node feature information. By encompassing multiple modalities, these approaches aim not only to create more generalizable models but also to enhance performance on specific tasks by leveraging complementary information across modalities [10, 18].

One such hybrid approach is GiGaMAE [18], which is a masked graph autoencoder. The authors identify a key challenge in developing such hybrid methods: the fundamentally different characteristics of graph topology and node attribute information. Graph topology is typically represented by a sparse, discrete adjacency matrix, whereas node features are continuous and high-dimensional. This disparity becomes especially problematic when constructing a unified learning objective. The authors argue against simply combining two separate objectives, as this can create gradient conflicts that degrade performance. To address this issue, the method implements the following:

- **Homogeneous latent space:** To avoid the difficulty of reconstructing fundamentally different data types, GiGaMAE maps all information into a unified latent space. Instead of targeting raw edges or features, the model uses embeddings from external models as reconstruction targets. For structural knowledge, the authors use node2vec [2] because it captures comprehensive topological information more efficiently than complex contrastive models. For attribute knowledge, they apply PCA [1] to the feature matrix to preserve useful information while removing noise. For hybrid knowledge, targets generated by existing GAEs or GCL can be used, though they are more computationally expensive. The model applies a weight setting strategy that prioritizes specific targets based on how a node was masked; for example, structural targets are prioritized for nodes with masked edges.
- **Mutual information based reconstruction loss:** The framework uses a reconstruction loss based on the Infomax principle to effectively integrate multiple targets. This approach allows the model to distinguish between the unique knowledge of a single target and the common knowledge shared between multiple sources. Because direct computation of mutual information is difficult, GiGaMAE uses the InfoNCE estimator to provide a lower bound for optimization. The training objective encourages the encoder to prioritize shared information, as this common knowledge is more generalizable across different downstream tasks.

Figure 3 illustrates the architecture of GiGaMAE. Unlike earlier methods, GiGaMAE masks both nodes and edges, as masking only one modality would hinder learning from the other [18]. Following encoding, the graph undergoes random remasking, which is applied to both edges and nodes. For reconstruction, the method employs separate MLP projectors for each target modality.

Another prominent method implementing a hybrid reconstruction approach is SeeGera [10], a masked autoencoder that, similar to GiGaMAE, masks both nodes and edges of the original graph before encoding. Additionally, it functions as a variational autoencoder, adopting semi-implicit variational inference to capture more complex latent probability distributions beyond the simple Gaussian distributions used in VGAE.

The method implements a dual-pathway architecture. The first pathway focuses on reconstructing the adjacency matrix from the latent representations of nodes. This pathway operates similarly to VGAE, employing a GCN encoder and an inner product decoder, though it differs in its approach to latent probability distributions, as mentioned above.

The second pathway handles feature reconstruction by producing latent representations for individual features. To infer the distribution, the framework utilizes an MLP that takes as input both the feature-node matrix and the latent node distribution produced by the adjacency reconstruction pathway, thereby accounting for correlations between nodes and features. This pathway then reconstructs node features by combining the latent representations of nodes with a weighted average of feature embeddings via a GCN decoder.

4 Benchmark Results

The following section provides a collection of benchmark results for the discussed graph autoencoder methods. The benchmarks

are conducted in two settings: an in-domain setting, where pre-training, fine-tuning, and evaluation are performed on the same dataset, and a transfer learning setting, where the model is pre-trained on one dataset and subsequently fine-tuned and evaluated on another. The in-domain benchmarks are sourced from Zhao et al. [31], and the transfer learning benchmarks are sourced from Song et al. [19]. In the in-domain setting, models are evaluated on both link prediction and node classification to assess their generalization across different tasks.

This section focuses exclusively on the experimental setup and empirical results. For an interpretation and analysis of these findings, please refer to Section 5.

4.1 In-Domain Representation Learning

For the in-domain benchmarks, the models are assessed on several widely-used graph benchmarks, including the Cora, CiteSeer, and PubMed citation networks [16], the Amazon Photo and Computers co-purchase networks [17], the Coauthor CS and Physics networks [17], and the ogbn-arxiv dataset from the Open Graph Benchmark [5]. In addition to the graph autoencoder benchmark results, performance metrics for state-of-the-art contrastive learning methods are reported, specifically GCA [33], CCA-SSG [28], and COSTA [30], to provide a more comprehensive comparison.

4.1.1 Link Prediction. The benchmark results for the link prediction task are sourced from the Bandana paper [31].

The edge sets are partitioned into training, validation, and test sets using a fixed 85/5/10% split. Notably, the evaluation employs dot-product probing, defined as:

$$A_{recon} = \sigma(ZZ^T) \quad (4)$$

This differs from the approach in most earlier structural reconstruction papers, which utilized a simple end-to-end method without probing or fine-tuning [31]. The authors argue that this decoupling of pre-training from the downstream task prevents overly optimistic evaluation results. Furthermore, it ensures that the results reflect the encoder’s inherent ability to embed graph topology without relying heavily on the MLP decoder.

The results can be seen in Table 1. Unfortunately, no link prediction benchmarks using this evaluation protocol were found for GiGaMAE and AUG-MAE; therefore, they are not included in the table. In the benchmarks conducted in the GiGaMAE paper, GiGaMAE performs comparably to S2GAE [18]. The AUG-MAE paper does not include link prediction benchmarks, as it is not designed for link prediction.

4.1.2 Node Classification. The node classification benchmark results are also sourced from Bandana [31], utilizing a standard linear evaluation protocol. Under this framework, the encoder is first pre-trained in an unsupervised manner, after which the decoder is discarded and the encoder weights are frozen. A logistic regression classifier is then trained on top of these fixed latent representations to produce the final classification logits.

Table 2 displays the results, reporting the Micro-F1 and Macro-F1 metrics. Although GiGaMAE and AUG-MAE were not included in the original Bandana benchmarks, results for AUG-MAE are provided here because it shares the same linear evaluation protocol and some datasets. However, only its Micro-F1 score is available. GiGaMAE’s results are omitted from the table because the model utilizes a different training and testing split, making its performance figures incomparable to the other baselines.

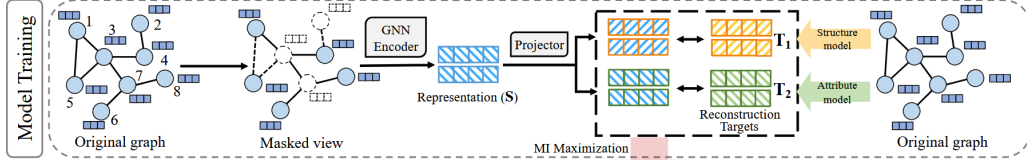


Figure 3: The GiGaMAE-Framework. Source: Shi et al. [18]

Table 1: AUC(%) and AP(%) results of link prediction. "OOM" stands for Out-Of-Memory. The "Avg Rank" column represents the global average rank across all datasets and both metrics. The best results are in bold.

Category	Year	Model	Cora	CiteSeer	PubMed	Photo	Computers	CS	Physics	Avg Rank
Contrastive Approaches	2021	GCA [33]	81.91±0.76 80.51±0.71	84.72±0.28 81.57±0.22	94.33±0.67 93.13±0.62	89.61±1.46 86.53±3.00	90.67±0.30 90.50±0.63	88.05±0.00 94.94±0.37	OOM	7.1
	2021	CCA-SSG [28]	67.54±1.30 72.74±1.18	78.88±2.73 77.42±4.56	74.97±0.28 77.11±0.26	91.04±2.98 89.68±3.85	83.85±1.35 84.04±1.74	83.54±0.98 78.66±1.06	77.40±0.08 73.33±0.10	8.9
Structure Reconstruction	2016	GAE [8]	94.66±0.26 94.22±0.39	95.19±0.45 95.70±0.31	94.58±1.12 94.26±1.65	71.45±0.95 65.99±0.96	70.99±1.03 67.88±0.82	93.78±0.36 89.87±0.59	88.88±1.11 82.45±1.59	7.1
	2016	VGAE [8]	91.24±0.48 92.27±0.43	94.55±0.48 95.34±0.37	95.46±0.04 94.29±0.07	95.61±0.05 94.63±0.06	92.69±0.03 88.27±0.08	87.34±0.43 80.24±0.55	89.27±0.83 82.79±1.14	5.4
	2023	S2GAE [20]	89.27±0.33 89.78±0.22	86.35±0.42 87.38±0.29	89.53±0.23 88.68±0.33	86.80±2.85 80.56±3.74	84.16±4.82 78.13±6.58	86.60±1.06 82.93±1.63	88.92±1.24 88.20±1.34	7.4
	2023	MaskGAE-e [9]	95.66±0.16 94.65±0.24	97.02±0.27 96.89±0.45	96.51±0.82 96.08±0.68	81.12±0.45 77.11±0.40	76.23±3.13 71.71±2.90	92.41±0.44 87.16±0.69	91.94±0.37 86.33±0.55	4.6
	2023	MaskGAE-p [9]	95.47±0.25 94.64±0.25	97.21±0.17 97.02±0.32	97.19±0.18 96.69±0.19	80.46±0.34 76.56±0.55	73.24±1.26 70.94±1.26	87.96±0.44 80.84±0.58	86.19±0.36 78.55±0.45	5.6
	2024	Bandana [31]	95.71±0.12 95.25±0.16	96.89±0.21 97.16±0.17	97.26±0.16 96.74±0.38	97.24±0.11 96.79±0.15	97.33±0.06 96.91±0.09	97.42±0.08 97.09±0.15	97.02±0.04 96.67±0.05	1.2
	2022	GraphMAE [4]	93.02±0.53 91.40±0.59	95.21±0.47 94.42±0.67	87.54±1.06 86.93±1.01	75.08±1.24 70.04±1.12	71.27±0.89 66.84±1.10	92.45±4.18 91.67±4.17	85.03±7.16 82.46±9.33	7.9
Feature Reconstruction	2023	GraphMAE2 [3]	93.26±1.00 91.65±0.98	95.26±0.14 94.36±0.20	90.85±0.91 90.37±0.92	73.03±2.24 68.77±1.50	72.20±2.09 67.97±1.52	94.57±0.32 92.76±0.54	94.56±0.81 93.86±1.09	6.4
Hybrid	2023	SeeGera [10]	95.49±0.70 95.90±0.64	94.61±1.05 96.40±0.89	95.19±3.94 94.60±4.17	95.25±1.19 94.04±1.18	96.53±0.16 96.33±0.16	95.73±0.70 93.17±0.53	OOM	4.2

4.1.3 Critical Discussion. The benchmarking results are subject to some notable limitations. First, the datasets used are relatively small, which may not fully represent the structural complexities or performance requirements of large-scale graphs. Second, the benchmarks are primarily conducted on homophilic datasets where connected nodes share similar features. This leaves the model’s effectiveness on heterophilic graphs, where edges connect dissimilar nodes, unexamined.

4.2 Transfer Learning

While most GAEs are evaluated extensively in an in-domain setting, domain transfer, where a model is pre-trained on one dataset and evaluated on another, is relatively uncommon. This is particularly true for GAEs that utilize a structure reconstruction training objective. To address this, Song et al. [19] introduced a systematic benchmark to evaluate the transferability of graph self-supervised learning methods across different datasets.

The paper benchmarks VGAE, representing Graph Autoencoders with a structure reconstruction objective; GraphMAE, representing GAEs with a feature reconstruction objective; and DGI [21] and GRACE [32] as representatives of contrastive methods.

The models are pre-trained on the large-scale ogbn-papers100M dataset [5]. They are then evaluated on five datasets from the same domain with structural and semantic similarities (Cora, Cite-seer, Pubmed, WikiCS, and DBLP), as well as three cross-domain datasets (A-Ratings, Child, and Photo) that differ substantially from the pre-training graph in both node features and structural characteristics. To handle feature heterogeneity between different graph datasets, the paper leverages LLMs to encode text-based node features into a shared semantic embedding space using SentenceBERT [15].

This paper focuses on transfer learning results for node classification. While the benchmark also evaluated link prediction, pre-training often led to negative transfer in that task, particularly when transferring to cross-domain datasets or using specific architectures like VGAE. For a more comprehensive analysis of the link prediction results and the exact experimental setup, please refer to [19].

4.2.1 Node Classification. For pre-training, all methods are trained on the source dataset for a maximum of 5 epochs. The best model checkpoint is selected based on the highest mean node classification performance across the downstream datasets.

Table 2: Micro-F1(%) and Macro-F1(%) of node classification. Best results are in bold. "OOM" stands for Out-Of-Memory. The "Avg Rank" column represents the average rank across all datasets and metrics.

Category	Year	Model	Cora	CiteSeer	PubMed	Photo	Computers	CS	Physics	ogbn-arxiv	Avg Rank
Contrastive Approaches	2021	GCA [33]	81.92±0.17 80.76±0.35	71.60±0.27 68.79±0.37	84.08±0.16 83.70±0.29	92.39±0.20 91.17±0.30	87.14±0.15 85.10±0.31	92.61±0.06 90.64±0.16	OOM	OOM	6.6
	2022	COSTA [30]	84.60±0.20 82.50±0.21	72.57±0.31 66.11±0.29	83.76±0.03 83.16±0.02	90.98±0.00 88.22±0.00	87.35±0.08 85.99±0.13	92.48±0.05 89.32±0.08	95.31±0.04 93.90±0.05	OOM	5.9
	2021	CCA-SSG [28]	83.96±0.38 83.01±0.48	73.45±0.44 68.75±0.51	81.81±0.53 81.31±0.51	92.87±0.36 91.69±0.49	88.61±0.29 87.24±0.52	93.01±0.29 90.73±0.51	95.31±0.07 93.76±0.10	69.52±0.09 47.39±0.51	4.4
Structure Reconstruction	2016	GAE [8]	80.15±0.34 78.44±0.90	69.79±0.36 62.64±0.58	80.51±0.53 79.62±0.38	91.07±0.09 89.83±0.13	87.92±0.12 86.02±0.25	90.46±0.29 89.75±0.32	93.04±0.03 91.02±0.04	69.58±0.32 48.25±0.53	9.2
	2016	VGAE [8]	76.30±0.49 74.70±0.60	58.85±0.79 52.80±0.91	75.73±0.22 75.39±0.24	89.58±0.20 86.61±0.40	84.99±0.19 82.26±0.30	92.33±0.07 89.09±0.16	94.40±0.07 92.28±0.11	69.94±0.30 47.05±0.79	10.7
	2023	S2GAE [20]	78.34±0.96 77.44±0.86	65.31±0.64 62.54±0.64	80.11±0.52 79.04±0.47	91.43±0.07 90.47±0.15	85.31±0.07 81.48±0.18	90.47±0.07 87.69±0.11	93.98±0.06 91.95±0.08	67.77±0.36 36.41±0.24	10.1
	2023	MaskGAE-e [9]	83.33±0.15 82.60±0.24	72.02±0.46 66.36±0.63	82.33±0.39 81.63±0.41	93.28±0.08 92.04±0.08	89.42±0.15 88.00±0.14	92.29±0.25 90.17±0.34	95.10±0.04 93.48±0.04	70.95±0.29 49.37±0.45	5.6
	2023	MaskGAE-p [9]	82.54±0.16 81.84±0.26	72.32±0.39 65.77±0.40	82.80±0.22 82.23±0.23	93.29±0.10 92.16±0.17	89.40±0.10 87.69±0.15	92.54±0.21 90.25±0.31	95.15±0.11 93.51±0.03	71.22±0.40 49.99±0.54	5.3
	2024	Bandana [31]	84.62±0.37 82.97±0.92	73.60±0.16 68.11±0.48	83.53±0.51 82.99±0.40	93.44±0.11 92.26±0.04	89.62±0.09 87.79±0.20	93.10±0.05 91.02±0.13	95.57±0.04 94.20±0.05	71.09±0.24 49.66±0.50	2.0
Feature Reconstruction	2022	GraphMAE [4]	84.05±0.59 83.07±0.53	73.06±0.37 67.78±0.85	80.98±0.47 80.26±0.48	92.92±0.40 91.93±0.47	89.24±0.45 88.12±0.78	93.09±0.14 91.42±0.15	95.65±0.07 94.19±0.09	71.30±0.24 51.15±0.15	3.6
	2023	GraphMAE2 [3]	83.84±0.54 82.80±0.46	73.48±0.34 68.70±0.42	81.34±0.44 80.68±0.43	93.30±0.20 92.19±0.24	89.01±1.53 87.63±1.79	91.31±0.07 88.89±0.13	95.25±0.05 93.78±0.07	71.82±0.00 50.42±0.00	5.1
	2024	AUG-MAE [23]	84.3±0.4 -	73.2±0.4 -	81.4±0.4 -	- -	- -	- -	- -	71.9±0.2 -	-
Hybrid	2023	SeeGera [10]	83.95±0.55 82.88±0.66	72.11±1.26 68.48±0.86	79.55±0.29 78.36±0.44	90.13±0.57 87.76±1.01	88.39±0.26 -	88.79±0.93 85.38±1.62	OOM	OOM	9.2

To adapt to downstream datasets, the paper tested three strategies: linear probing, fine-tuning, and in-context learning. Fine-tuning the pre-trained GNN encoders did not show significant performance differences compared to random initialization. While in-context learning showed some gains, linear probing generally performed better, was simpler, and more robust. Consequently, this section only discusses the results using linear probing.

In linear probing, the encoder is frozen after the pre-training and a linear classifier is trained to classify the embeddings produced by the encoder. The linear classifier for the downstream task was trained using a few-shot protocol with 5 training nodes per class, 500 validation nodes, and the remaining nodes for testing. Testing was conducted over five random splits, and the mean accuracy (ACC) is reported in Table 3. The results using both GCN and GAT encoders are reported.

4.2.2 Critical Discussion. Despite its contributions, the benchmark has several limitations worth noting.

First, there is a notable strength in the random baselines. Randomly initialized GNNs achieve high performance primarily because the LLM-derived features already provide rich semantic information, leaving little room for improvement through pre-training. This is not necessarily a disadvantage, but should be taken into consideration when interpreting the results.

Second, the benchmark is limited by its choice of feature encoding, as only SentenceBERT was used to unify the feature space. While this allows for a consistent comparison, the results may be specific to this encoder, and different language models might produce different transfer behaviors.

Finally, and most critically, the method-specific hyperparameters were adopted from the original papers rather than being optimized for the specific pre-training dataset and task. This makes a fair comparison between methods difficult. Nevertheless, the results demonstrate that cross-domain pre-training of GNNs with SSL can improve downstream performance, particularly for graph autoencoders.

5 Discussion

5.1 What to reconstruct?

A key aspect of comparison reflecting in the proposed taxonomy is the reconstruction target. The expressiveness of learned representations depends heavily on the choice of reconstruction objective [13]. This is also evidenced in the benchmark results. Methods based solely on node-feature reconstruction perform poorly on link prediction tasks (see Table 1). Conversely, recent structure-based methods such as Bandana [31] and MaskGAE [9] achieve surprisingly strong performance on node classification, rivaling purely feature-based approaches (see Table 2).

While showing very promising results, the diversity of possible reconstruction targets in structural approaches complicates the direct comparison. Different methods reconstruct fundamentally different structural properties: the adjacency matrix [8, 9, 20], node degrees [9], edge bandwidths [31], or node positions in spectral space [13]. Each target captures distinct aspects of graph structure, yet a systematic comparison of their relative informativeness remains an open question. In the author’s opinion a more rigorous framework for evaluating and comparing the

Table 3: Linear Probing Accuracy (%) for Node Classification. Best results are in bold. The top value in each cell represents a GCN encoder, while the bottom value represents the GAT encoder. Two average rank columns are provided for in-domain and cross-domain datasets, respectively.

Model	In-domain Datasets					Avg Rank	Cross-domain Datasets			Avg Rank
	Citeseer	Cora	DBLP	PubMed	WikiCS		A-Ratings	Child	Photo	
Rand. Init. GNN	68.0±1.7	73.7±1.6	71.2±3.3	66.8±3.3	61.8±6.0	3.6	25.7±4.1	20.8±5.1	51.2±1.8	3.3
	68.7±1.6	74.4±1.7	73.1±2.8	69.0±3.2	70.6±3.3	3.6	23.9±2.0	24.5±3.3	56.9±2.3	3.7
DGI [21]	66.3±2.4	73.7±1.7	69.2±4.6	67.2±2.0	60.1±5.3	4.2	20.7±6.2	26.2±4.6	49.5±1.9	4.0
	69.8±1.5	74.1±1.5	72.1±2.4	70.3±3.8	70.5±3.7	3.4	23.6±2.5	24.6±5.3	56.6±2.6	4.0
GRACE [32]	69.2±2.6	73.7±1.3	70.1±2.8	67.1±3.2	61.9±6.5	3.2	20.3±4.5	26.2±5.1	51.4±1.2	3.3
	67.0±1.8	74.4±2.3	72.0±3.7	67.8±2.2	70.8±3.6	4.2	23.9±4.1	24.0±2.6	57.1±2.6	3.7
VGAE [8]	69.7±1.8	74.1±2.0	70.0±2.5	67.4±2.6	61.0±7.3	2.6	22.7±4.7	35.3±2.2	52.5±2.1	2.0
	68.8±0.9	77.4±0.9	72.9±3.0	69.1±3.6	71.4±3.1	2.4	35.4±2.5	27.6±4.0	59.8±3.5	1.0
GraphMAE [4]	68.6±1.3	74.8±1.1	73.1±3.1	70.2±3.1	70.4±3.5	1.4	25.6±3.6	25.9±4.2	57.5±1.2	2.3
	69.7±1.6	76.2±1.5	73.9±2.7	71.7±3.1	72.6±3.2	1.4	26.3±3.6	24.4±4.1	58.8±2.0	2.7

expressiveness of these reconstruction targets could be highly useful for advancing the field.

Still, feature-based methods demonstrate that feature reconstruction alone can yield excellent results on node classification (see Table 2), suggesting that node-feature reconstruction remains valuable for learning informative representations. The hybrid approach SeeGera [10] shows competitive performance on node classification but lags behind recent structure-based methods like Bandana. GiGaMAE [18] achieves strong performance on both link prediction and node classification, indicating the potential of multi-target approaches.

Another important distinction concerns whether reconstruction targets raw features and edges or operates in a latent space. GiGaMAE performs reconstruction exclusively in latent space provided by external models. GraphMAE2 [3] employs latent representation reconstruction as an additional regularizer alongside raw node-feature matrix reconstruction. The primary advantage of latent reconstruction is its robustness to noise, which is very prevalent in graph data [3]. Features, in particular, are often noisy approximations generated by feature extractors or derived from noisy real-world measurements [3].

Looking forward, the future likely lies in hybrid approaches that effectively utilize multiple modalities and combine structural and feature-based targets to capture complementary aspects of graph data. However, current hybrid methods face significant challenges. A fundamental difficulty is the unification of raw structural information and node features that have distinct characteristics [18]. Structural Information is usually discrete and sparse, while feature information is continuous and often high dimensional [18]. GiGaMAE approaches this problem by utilizing latent representations from external models. The main limitation of this approach is the heavy reliance on these external models. Addressing these challenges while leveraging the strengths of multi-target reconstruction remains a promising direction for future research.

5.2 Architectural Considerations

The choice of encoder and decoder architectures significantly influences the quality of the learned representations in graph self-supervised learning.

Different reconstruction objectives favor different encoder architectures. Node-feature reconstruction methods typically employ GAT encoders for node classification tasks and GIN encoders for graph classification [3, 4, 23]. In contrast, structure reconstruction frameworks commonly utilize GCNs [8]. However, existing standard GNNs may lack sufficient expressiveness for certain reconstruction tasks [20]. To address this limitation, methods such as S2GAE [20] and GraphPAE [13] enhance their GNN encoders by extracting additional structural information beyond standard message-passing schemes.

GraphMAE demonstrates that employing a GNN decoder rather than a simple MLP decoder facilitates learning more structurally informative representations. This choice shifts the reconstruction task from individual node recovery to neighborhood-based recovery, which is particularly important given that feature vectors in practice are often weakly informative, making MLPs potentially insufficient for effective reconstruction [4]. However, a powerful decoder introduces the risk of a "lazy" encoder that relies too heavily on the decoder's capacity. To mitigate this issue, regularization strategies are essential. GraphMAE employs random remasking during training, while GraphMAE2 introduces an additional latent reconstruction target to encourage the encoder to learn more robust representations.

5.3 Evolution of Masking Strategies

Early structural graph masked autoencoder methods predominantly adopt discrete masking approaches [9, 20]. However, this strategy may be suboptimal as it blocks message flows and destroys graph connectivity. To address this limitation, Bandana [31] introduces non-discrete bandwidth masking, which treats edges as continuous channels rather than binary on/off states, thereby preserving information flow throughout the graph.

Building on GraphMAE, both AUG-MAE [23] and StructMAE [11] argue that random masking is fundamentally suboptimal. AUG-MAE offers proof, that masked node-feature reconstruction methods such as GraphMAE implicitly perform graph contrastive learning. From this perspective, AUG-MAE approaches masking through the lens of alignment, which is a key property in contrastive graph learning. Random masking tends to sample a large number of easy-to-align positive pairs

but only a few hard-to-align ones, resulting in poor performance on challenging cases [23]. To address this, AUG-MAE proposes an easy-to-hard adversarial masking strategy that progressively increases difficulty. Similarly, StructMAE introduces structure-based scoring to guide the masking process, specifically targeting the weakness of feature reconstruction methods in capturing structural information [11]. Ablation studies demonstrate that these adaptive masking strategies yield significant performance gains on both node and graph classification tasks.

The recent paper GraphPAE [13] evaluates discrete masking strategies from a spectral perspective, revealing a critical limitation: existing masked graph autoencoders act primarily as low-pass filters, overlooking high-frequency information. Specifically, simple discrete masking of nodes or edges creates discrepancies predominantly in the low-frequency bands between masked and original graphs, leading these methods to primarily reconstruct low-frequency components while neglecting high-frequency signals (see Figure 4). This bias results in particularly poor performance on heterophilic graphs.

These findings suggest that improved masking strategies remain a promising and active research direction, with significant potential for further performance improvements.

5.4 Further Challenges and Outlook

Despite the recent advances, the pursuit of more general and robust models still faces several fundamental challenges.

A primary concern is hyperparameter sensitivity. Current methods often rely on numerous dataset-dependent hyperparameters that require careful tuning to optimize performance. These include the masking ratio [3, 4, 9, 18, 20, 23, 31], scaled cosine error coefficients [3, 4, 23], and target-specific weights such as those used in GiGaMAE, where different weights are assigned to different reconstruction targets depending on node categories. This sensitivity not only complicates model deployment but also limits the generalizability of these methods across diverse graph datasets and application domains.

Another big hurdle on the path to more general models is limited scalability. In the ablation study GraphMAE2 demonstrates two critical challenges regarding scaling with existing GNNs [3]: increasing network depth leads to oversmoothing, where node representations become indistinguishable, while increasing network width yields diminishing returns while increasing computation.

In the author’s opinion another fundamental challenge is the lack of comprehensive theoretical understanding. While theoretical equivalence between masked structural reconstruction and masked node-feature reconstruction methods has been established [9, 23], many recent papers primarily offer empirical motivation and experimental results without deeper theoretical analysis of why and when these methods succeed. Greater effort toward developing a unified theory for generative graph pre-training could reveal additional pathways for improvement and provide guidance for method design, moving the field beyond trial-and-error approaches.

5.5 Transferability

While the in-domain benchmarks demonstrate that recent GAEs generalize well across different tasks, they provide no insight into transferability across datasets. This section briefly interprets the transfer learning benchmarks and outlines how GAEs might

be leveraged to achieve more general and transferable representations.

The following discussion is mostly based on the benchmarks and their interpretation in Song et al. [19]. Despite the absence of method specific hyperparameter tuning, which is a critical limitation that hinders a fair comparison between methods, the benchmarks yield several notable insights:

- **LLM Preprocessing:** Using Large Language Models to construct a shared semantic space is shown to be a highly effective preprocessing step. It mitigates feature heterogeneity across diverse graph domains and produces inherently richer representations than shallow features, enabling even randomly initialized GNNs to achieve strong baseline performance [19].
- **Structure and Feature Reconstruction Both Aid Generalization:** Both GraphMAE, as a representative of node feature reconstruction, and VGAE, as a representative of structural reconstruction, yield performance gains over random initialization on most datasets under linear probing (see Table 3). Although modest, these gains confirm that graph autoencoder frameworks are viable in cross-domain settings. By contrast, the tested contrastive methods perform on par with or worse than random initialization across domains. While VGAE marginally outperforms GraphMAE in the cross-domain datasets on the node classification benchmarks, Song et al. [19] identify GraphMAE as more consistent and stable: it achieves gains over random initialization across all tasks and adaptation strategies, which VGAE does not, and exhibits superior training stability. This suggests that feature reconstruction may be particularly well-suited for learning general, transferable representations [19].
- **GNN Architecture as a Potential Bottleneck:** GAT encoders yield stronger performance in the node classification setting than GCNs (see Table 3), yet are especially prone to collapse under fine-tuning, showing no improvement over random initialization [19]. The failure of fine-tuning across all tested architectures indicates that standard GNNs are presently limited in their ability to consolidate and scale knowledge acquired during large-scale pre-training. The fact that only linear probing yields consistent results restricts the applicability of GAEs for transfer learning to scenarios with extreme label-scarcity, such as few-shot learning [19].

To summarize, GAEs appear to facilitate the learning of transferable representations but currently suffer from limited scalability and difficult adaptation to downstream tasks, which can be largely attributed to the limitations of existing GNN encoders. This heavily limits the applicability of the presented methods in a transfer learning setting.

To deal with this issue, a promising research direction is offered by the Generative Graph Pattern Machine (G^2PM) [26], which shares the masked autoencoder philosophy but addresses the encoder bottleneck by replacing traditional message-passing with a pure Transformer architecture. By modeling graph data as sequences rather than relying on neighborhood aggregation, this framework overcomes standard GNN constraints such as oversmoothing and over-squashing. Notably, G^2PM demonstrates a clear scaling law, where performance improves consistently as model size increases up to 60 million parameters. This shift

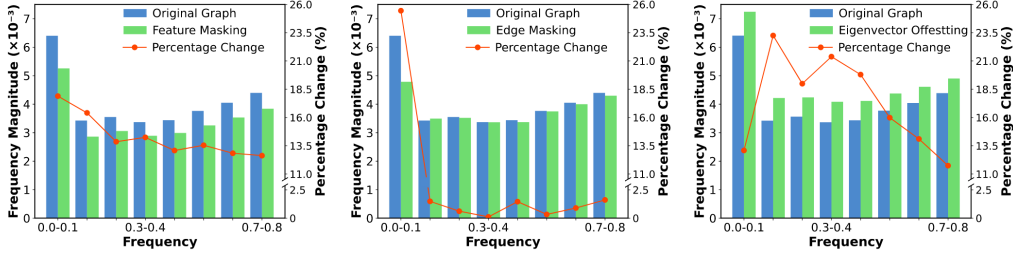


Figure 4: Frequency magnitudes of augmented and original graphs. The figure shows, that traditional discrete masking primarily leads to discrepancies in the low-frequency domain. Source: Liu et al. [13]

toward scalable, non-message-passing encoders is, in the author’s opinion, a very promising step toward extending the graph autoencoding framework to large-scale, general-purpose graph foundation models.

6 Conclusion

The results of this analysis show that the introduction of masked graph modeling has enabled graph autoencoders to become a worthy alternative to graph contrastive learning. These methods avoid many of the disadvantages inherent in graph contrastive learning, such as computationally expensive negative sampling and reliance on domain-specific data augmentation, while maintaining comparable or even superior performance on downstream tasks.

The in-domain evaluation shows that graph autoencoder methods achieve strong performance overall, with recent methods performing well in both link prediction and node classification, demonstrating good generalization ability across tasks. Nevertheless, several avenues for improvement have been identified. First, identifying the most informative reconstruction targets remains an open question, as different methods currently reconstruct fundamentally different properties. Furthermore, constructing approaches that incorporate multiple target modalities remains challenging, as unifying discrete structural information with continuous, high-dimensional node features proves difficult. Another interesting avenue of exploration, pursued by many recent works, concerns the masking strategies used in masked graph autoencoders. Discrete edge masking may negatively impact both local and global informativeness [31], while random node masking ignores the need to learn alignments between hard-to-align positive pairs [23]. Finally, there is a clear need for stronger theoretical foundations to move the field beyond empirical trial-and-error and provide a unified understanding of why these generative pre-training methods succeed.

While graph autoencoders show strong performance in the in-domain setting, transferring encoders pre-trained on one domain to another remains highly limited. Song et al. [19] shows that fine-tuning these pre-trained encoders yields no benefits over random initialization. This limitation is primarily attributed to the constraints of existing GNNs. Recent research [26] has explored applying the masked autoencoding paradigm to more powerful and scalable encoders that do not suffer from the scaling problems of traditional GNNs, which appears to be a promising direction on the path towards graph foundation models.

Limitations

This seminar paper has the following limitations:

- The absence of empirical results for graph classification tasks.
- The omission of heterophilic graphs as a challenging case study, including how they affect the performance of the presented methods.
- The collected benchmark results are subject to several notable limitations.

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A Supplementary Materials

A.1 Dataset Statistics

Table 4: Dataset statistics from the in-domain benchmarks [31].

Dataset	nodes	edges
Cora [16]	2,708	10,556
CiteSeer [16]	3,327	9,104
PubMed [16]	19,717	88,648
Photo [17]	7,487	119,043
Computers [17]	13,381	245,778
CS [17]	18,333	81,894
Physics [17]	34,493	247,962
WikiCS [14]	11,701	216,123
ogbn-arxiv [5]	169,343	2,315,598

Table 5: Dataset statistics from the transfer learning benchmarks [19]

Dataset	nodes	edges
Cora [16]	2,708	10,556
Citeseer [16]	3,186	8,450
WikiCS [14]	11,701	431,726
DBLP [19]	431,326	14,376
Pubmed [16]	19,717	88,648
Amazon Ratings [19]	186,100	24,492
Child [19]	76,875	2,325,044
Photo [17]	48,362	873,782